

Tugas Week 5
Jaringan Komputer



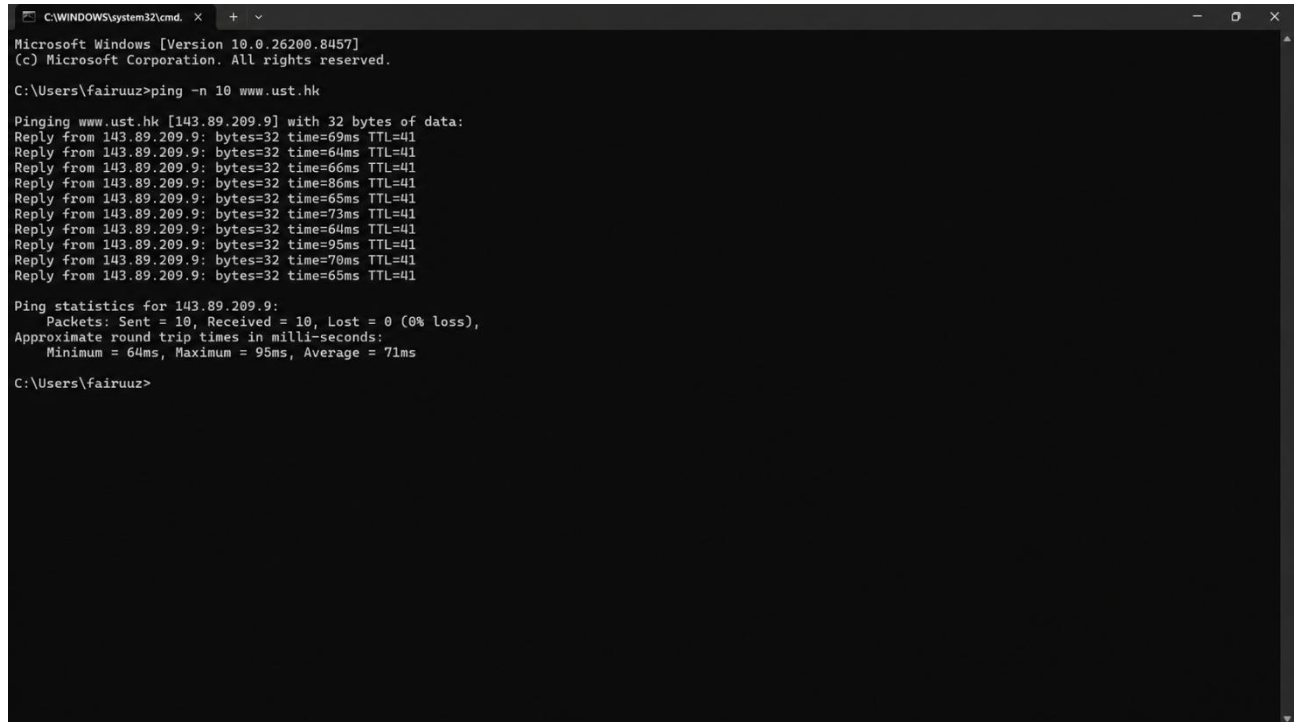
Nama Dan Nim:
Fathan Aditya Rachman, 103072400153

ICMP

1. Mahasiswa dapat menginvestigasi cara kerja protokol ICMP menggunakan Wireshark
2. Mahasiswa dapat membuat program ICMP Pinger

ICMP dan PING

1. **ping -n 10 www.ust.hk**



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz>ping -n 10 www.ust.hk

Pinging www.ust.hk [143.89.209.9] with 32 bytes of data:
Reply from 143.89.209.9: bytes=32 time=69ms TTL=41
Reply from 143.89.209.9: bytes=32 time=64ms TTL=41
Reply from 143.89.209.9: bytes=32 time=66ms TTL=41
Reply from 143.89.209.9: bytes=32 time=86ms TTL=41
Reply from 143.89.209.9: bytes=32 time=65ms TTL=41
Reply from 143.89.209.9: bytes=32 time=73ms TTL=41
Reply from 143.89.209.9: bytes=32 time=64ms TTL=41
Reply from 143.89.209.9: bytes=32 time=95ms TTL=41
Reply from 143.89.209.9: bytes=32 time=70ms TTL=41
Reply from 143.89.209.9: bytes=32 time=65ms TTL=41

Ping statistics for 143.89.209.9:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 64ms, Maximum = 95ms, Average = 71ms

C:\Users\fairuuz>
```

2. ICMP (Echo Ping Request)

The screenshot shows a Wireshark capture of ICMP Echo (ping) requests. The packet list on the left shows packets 260 through 428, all of which are Echo (ping) requests from 192.168.0.102 to 143.89.209.9. The packet details pane on the right shows the structure of an Internet Control Message Protocol (ICMP) Echo (ping) request. The request has a type of 8, a code of 0, a checksum of 0x4d59, an identifier of 1, and a sequence number of 2. The data field contains 32 bytes of data.

No.	Time	Source	Destination	Protocol	Length	Info
260	11.6241095..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 261)
261	11.6938367..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=41 (request in 260)
278	12.6332043..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 279)
279	12.6973911..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=41 (request in 278)
314	13.6381918..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 315)
315	13.7042120..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=41 (request in 314)
316	13.7042136..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=41
338	14.6497435..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 339)
339	14.7352278..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=41 (request in 338)
377	15.6601579..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 378)
378	15.7254200..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=41 (request in 377)
407	16.6749078..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 408)
408	16.7485676..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=41 (request in 407)
409	16.7499210..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=41
421	17.6935529..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 422)
422	17.7582762..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=41 (request in 421)
427	18.7013746..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=9/2304, ttl=128 (reply in 428)
428	18.7967377..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=9/2304, ttl=41 (request in 427)

Frame 260: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface
Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4)
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 143.89.209.9
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x4d59 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 2 (0x0002)
Sequence Number (LE): 512 (0x0200)
[Response frame: 261]
Data (32 bytes)

3. ICMP (Echo Ping Reply)

The screenshot shows a Wireshark capture of ICMP Echo (ping) replies. The packet list on the left shows packets 261 through 427, all of which are Echo (ping) replies from 143.89.209.9 to 192.168.0.102. The packet details pane on the right shows the structure of an Internet Control Message Protocol (ICMP) Echo (ping) reply. The reply has a type of 0, a code of 0, a checksum of 0x5559, an identifier of 1, and a sequence number of 2. The data field contains 32 bytes of data.

No.	Time	Source	Destination	Protocol	Length	Info
260	11.6241095..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 261)
261	11.6938367..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=41 (request in 260)
278	12.6332043..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 279)
279	12.6973911..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=41 (request in 278)
314	13.6381918..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 315)
315	13.7042120..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=41 (request in 314)
316	13.7042136..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=41
338	14.6497435..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 339)
339	14.7352278..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=41 (request in 338)
377	15.6601579..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 378)
378	15.7254200..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=41 (request in 377)
407	16.6749078..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 408)
408	16.7485676..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=41 (request in 407)
409	16.7499210..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=41
421	17.6935529..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 422)
422	17.7582762..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=41 (request in 421)
427	18.7013746..	192.168.0.102	143.89.209.9	ICMP	74	Echo (ping) request id=0x0001, seq=9/2304, ttl=128 (reply in 428)
428	18.7967377..	143.89.209.9	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=9/2304, ttl=41 (request in 427)

Frame 261: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface
Ethernet II, Src: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4), Dst: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d)
Internet Protocol Version 4, Src: 143.89.209.9, Dst: 192.168.0.102
Internet Control Message Protocol
Type: Echo (ping) reply (0)
Code: 0
Checksum: 0x5559 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 2 (0x0002)
Sequence Number (LE): 512 (0x0200)
[Request frame: 260]
[Response time: 69,727 ms]
Data (32 bytes)

ICMP dan Traceroute

1. c:\windows\system32\tracert www.inria.fr

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz>c:\windows\system32\tracert www.inria.fr

Tracing route to inria.fr [128.93.162.83]
over a maximum of 30 hops:

  1  33 ms  42 ms  6 ms  192.168.0.1
  2  18 ms  6 ms  4 ms  192.168.1.1
  3  24 ms  13 ms  21 ms  10.114.0.1
  4  26 ms  8 ms  5 ms  180.250.252.69
  5  *  41 ms  *  180.240.190.77
  6  53 ms  58 ms  *  180.240.190.77
  7  187 ms  179 ms  180 ms  180.240.192.233
  8  198 ms  193 ms  193 ms  180.240.196.1
  9  273 ms  297 ms  290 ms  renater-par.franceix.net [37.49.236.19]
 10  192 ms  192 ms  196 ms  hu0-4-0-1-ren-nr-orsay-rtr-091.noc.renater.fr [193.51.180.43]
 11  229 ms  209 ms  198 ms  inria-rocquencourt-vl1631-te1-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
 12  291 ms  283 ms  287 ms  unit240-reth1-vfw-ext-dcl.inria.fr [192.93.122.19]
 13  193 ms  197 ms  193 ms  prod-inriafr-cms.inria.fr [128.93.162.83]

Trace complete.

C:\Users\fairuuz>
```

2. ICMP after c:\windows\system32\tracert www.inria.fr in CMD

The Wireshark capture shows a sequence of ICMP Echo (ping) requests and responses. The packet details pane shows the structure of an ICMP Echo request:

- Frame 85: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface
- Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPlink 5b:1b:b4 (bc:07:1a:53:08:00)
- Internet Protocol Version 4, Src: 192.168.0.102, Dst: 128.93.162.83
- Internet Control Message Protocol

The packet bytes pane shows the raw data of the ICMP Echo request:

```
0000  bc 07 1d 5b 1b b4 24 b2  b9 af cf 7d 08 00 45 00  ...[. $ ...] E
0010  00 5c 6b 5c 00 00 01 01  6a 86 c0 a8 00 66 80 5d  \K.... j...f.
0020  a2 53 08 00 f7 cb 00 01  00 33 00 00 00 00 00 00  S.....3.....
0030  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  .....
0040  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  .....
0050  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  .....
0060  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  .....
```

3. Detail Time-To-Live Exceeded

The screenshot shows a Wireshark packet capture of ICMP Echo (ping) requests and responses. The packet list pane displays the following entries:

No.	Time	Source	Destination	Protocol	Length	Info
85	12.2340709..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=51/13056, ttl=1 (no response found!)
86	12.2445549..	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
87	12.2452789..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=52/13312, ttl=1 (no response found!)
88	12.2464346..	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
89	12.2468882..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=53/13568, ttl=1 (no response found!)
90	12.2485626..	192.168.0.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
100	12.6682478..	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
106	14.1807911..	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
123	15.7009131..	192.168.0.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
139	18.2256972..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=54/13824, ttl=2 (no response found!)
139	18.2401958..	192.168.1.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
140	18.2477031..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=55/14080, ttl=2 (no response found!)
142	18.2530041..	192.168.1.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
143	18.2533691..	192.168.0.102	128.93.162.83	ICMP	106	Echo (ping) request id=0x0001, seq=56/14336, ttl=2 (no response found!)
144	18.2579009..	192.168.1.1	192.168.0.102	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
148	18.2756217..	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
151	19.7698834..	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)
157	21.2844177..	192.168.1.1	192.168.0.102	ICMP	120	Destination unreachable (Port unreachable)

The packet details pane shows the structure of an ICMP Echo request (Frame 86):

- Frame 86: Packet, 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface 0
- Ethernet II, Src: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4), Dst: LiteonTechno_af:cf:7d (24:b2:b0:00:00:00)
- Internet Protocol Version 4, Src: 192.168.0.1, Dst: 128.93.162.83
- Internet Control Message Protocol
 - Type: Time-to-live exceeded (11)
 - Code: 0 (Time to live exceeded in transit)
 - Checksum: 0xf4ff [correct]
 - [Checksum Status: Good]
 - Unused: 00000000
 - Internet Protocol Version 4, Src: 192.168.0.102, Dst: 128.93.162.83
 - Internet Control Message Protocol

The packet bytes pane shows the raw data of the packet, including the ICMP Echo request and the Time-to-live exceeded message.